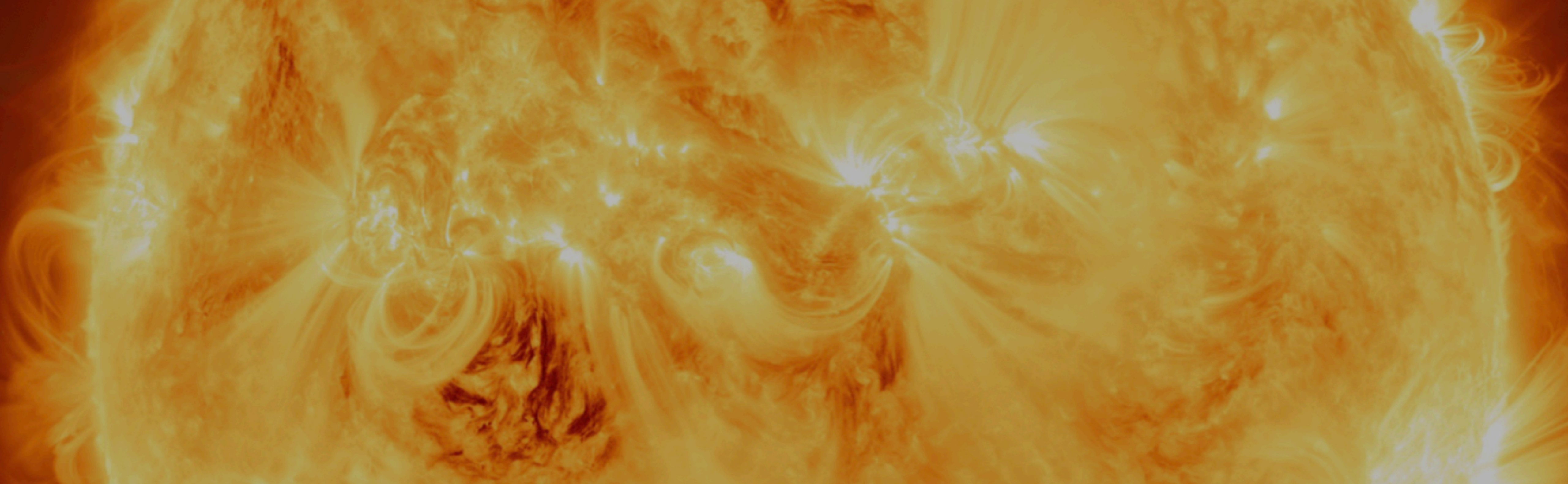


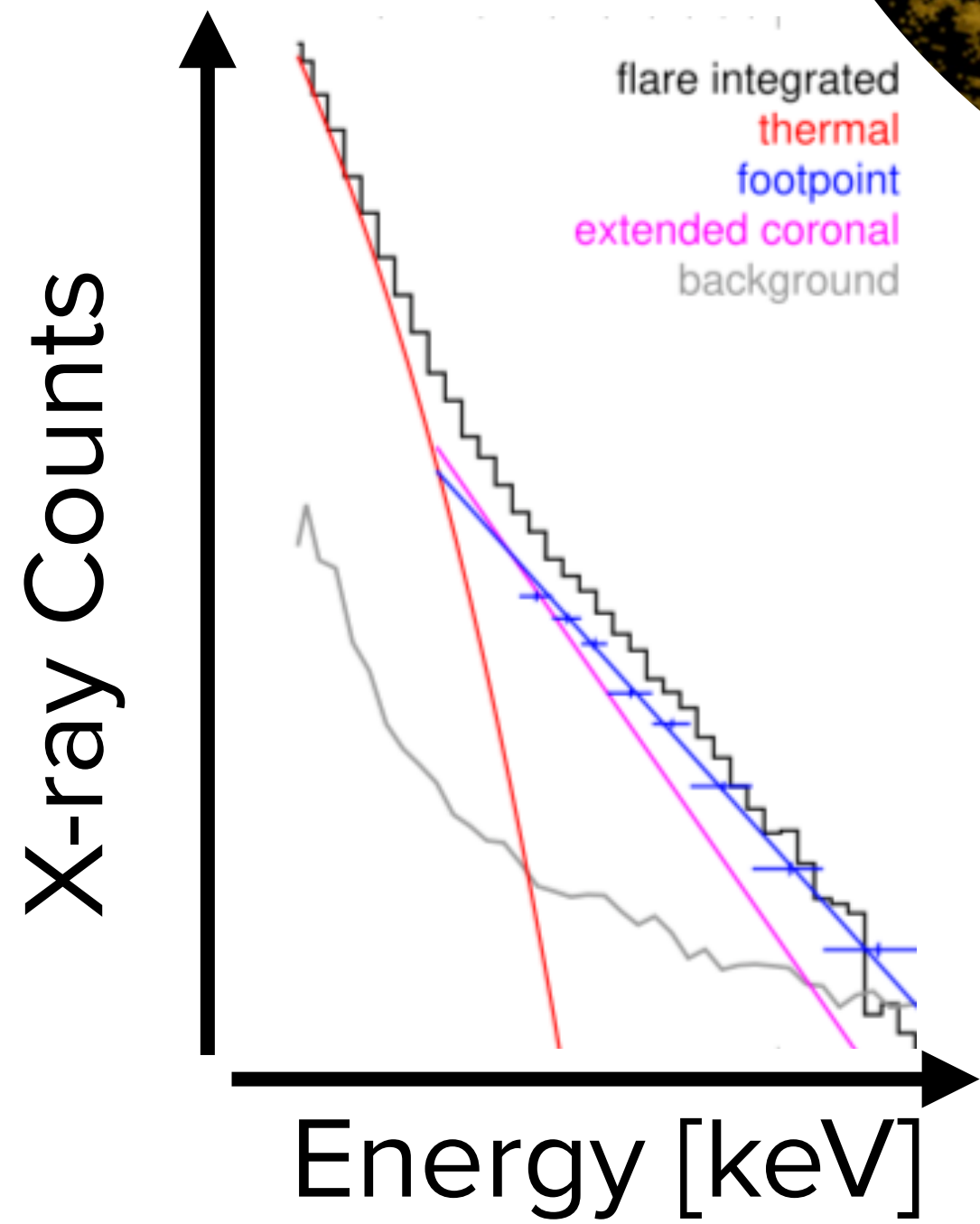
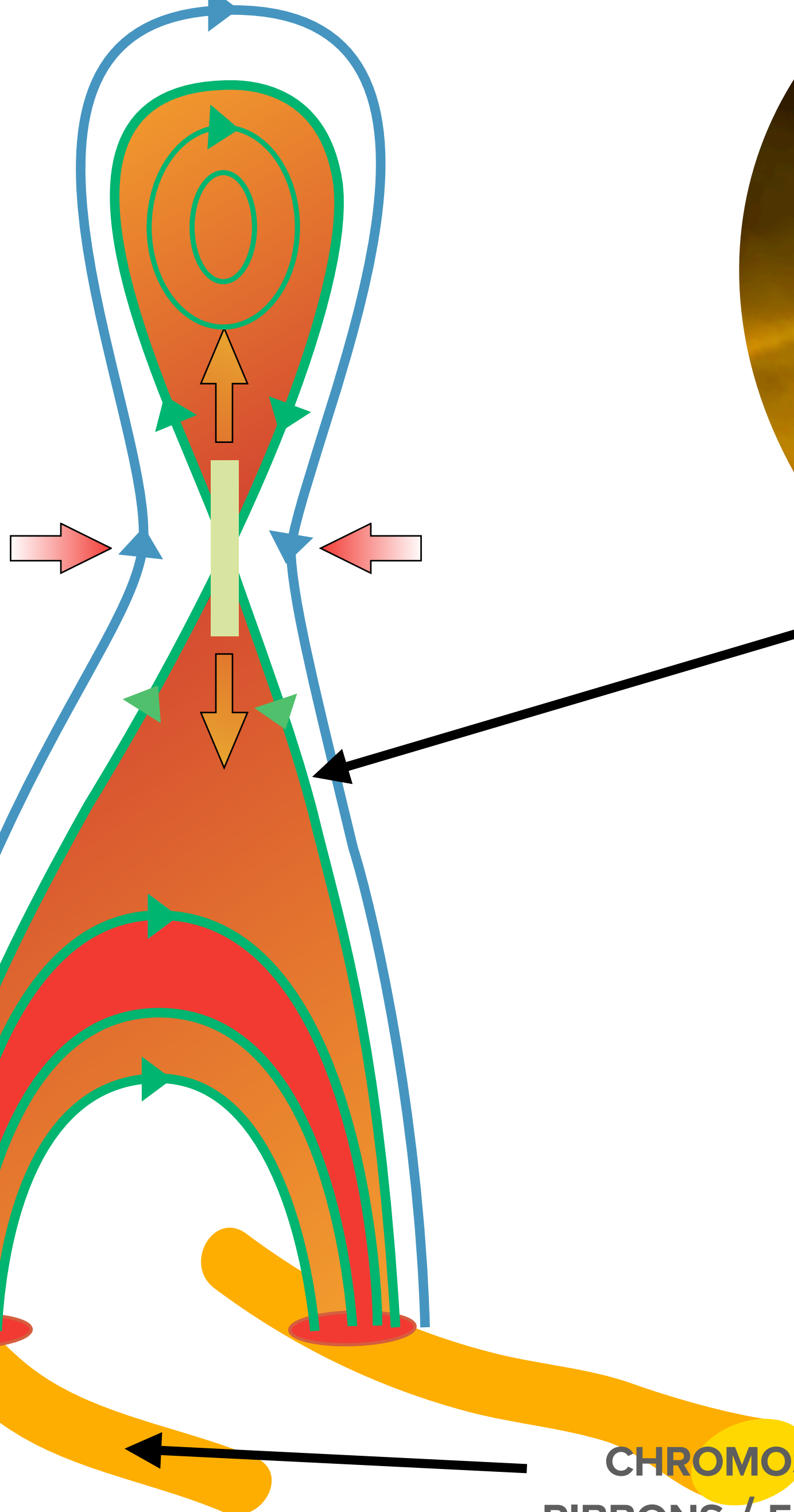
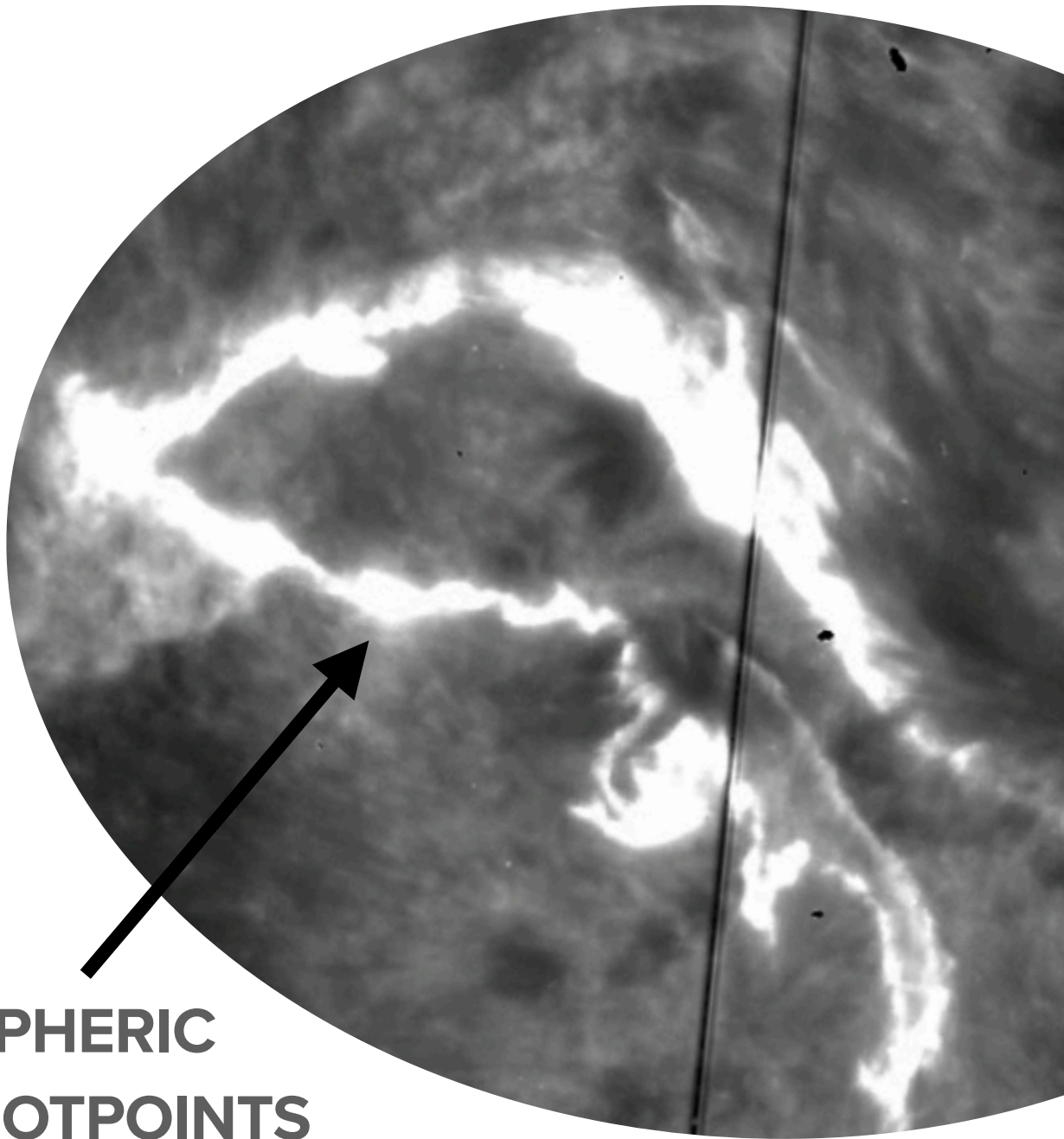
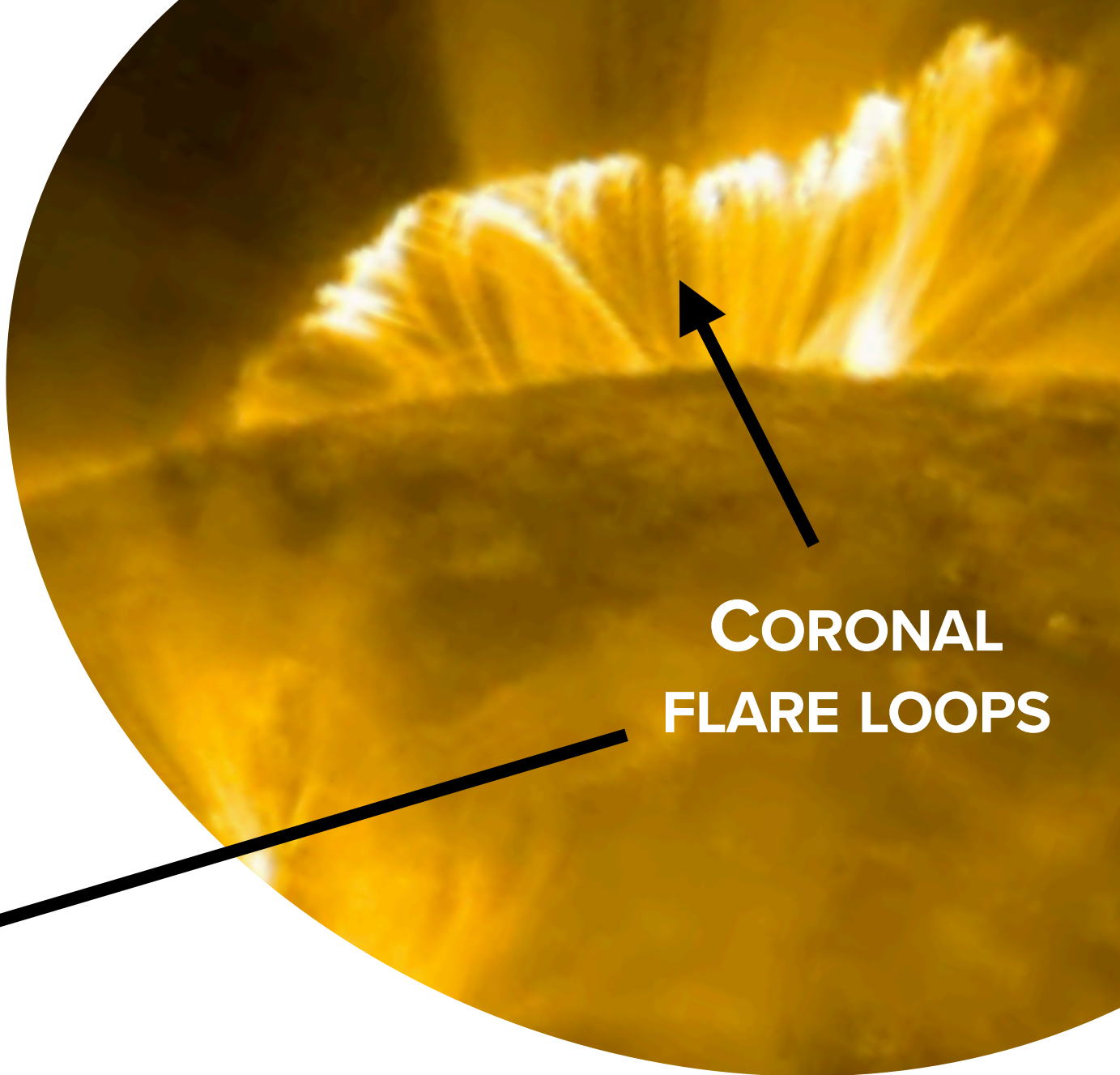
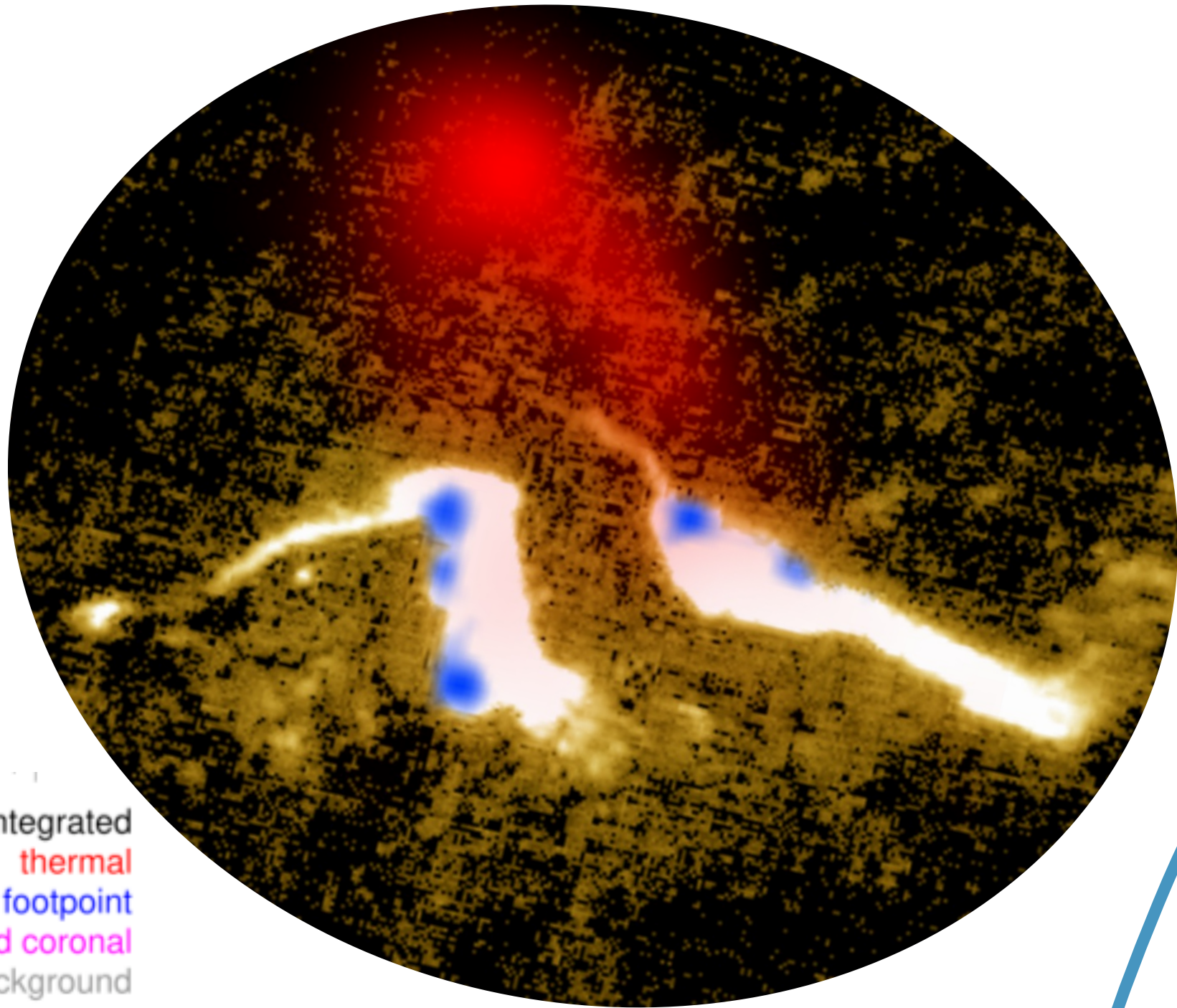


NON-THERMAL ENERGY FLUX DENSITY IN SOLAR FLARES

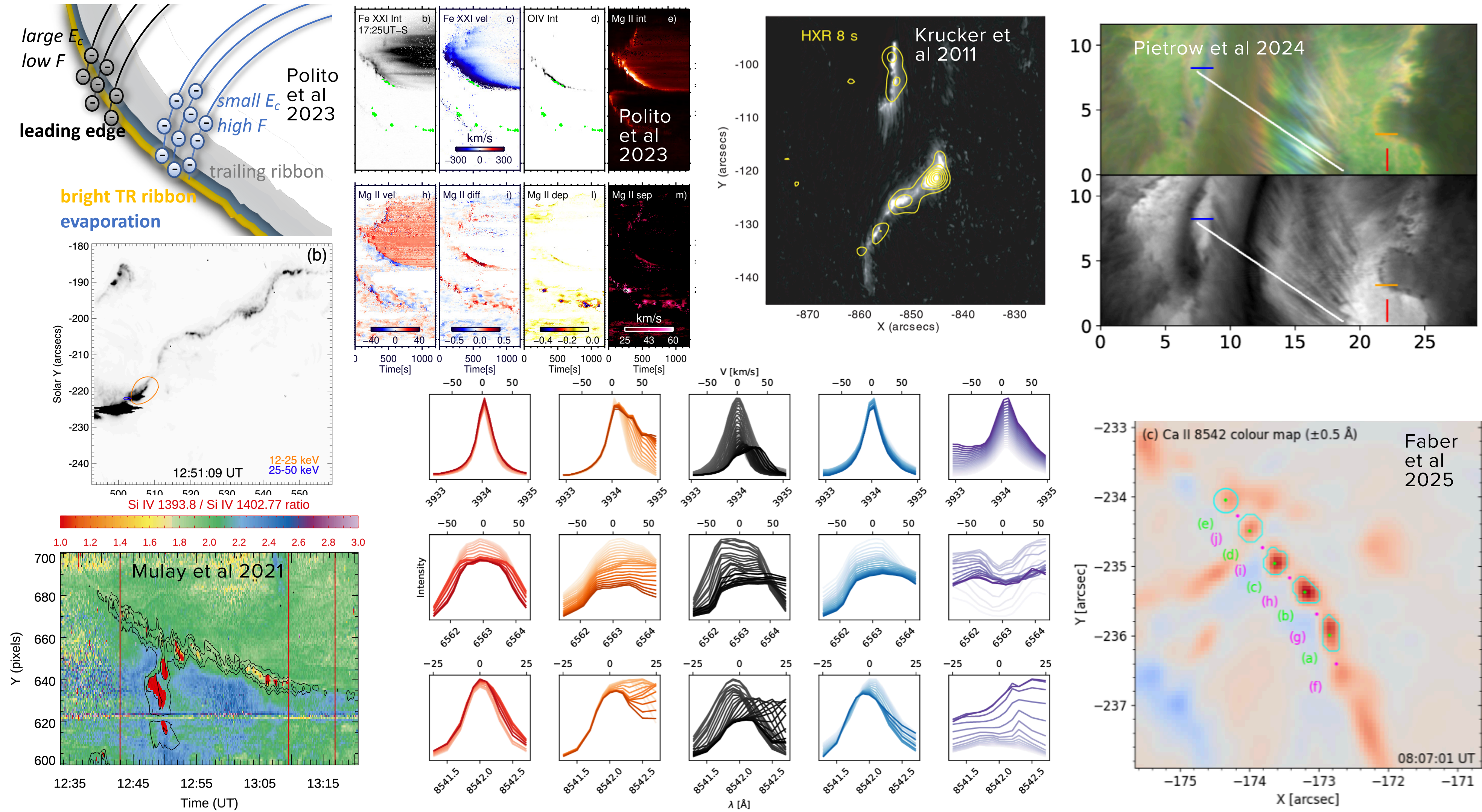
Dr. Graham S. Kerr

With: Adam F. Kowalski, Iain G. Hannah, Lyndsay Fletcher & Joel C. Allred

SOLAR FLARES



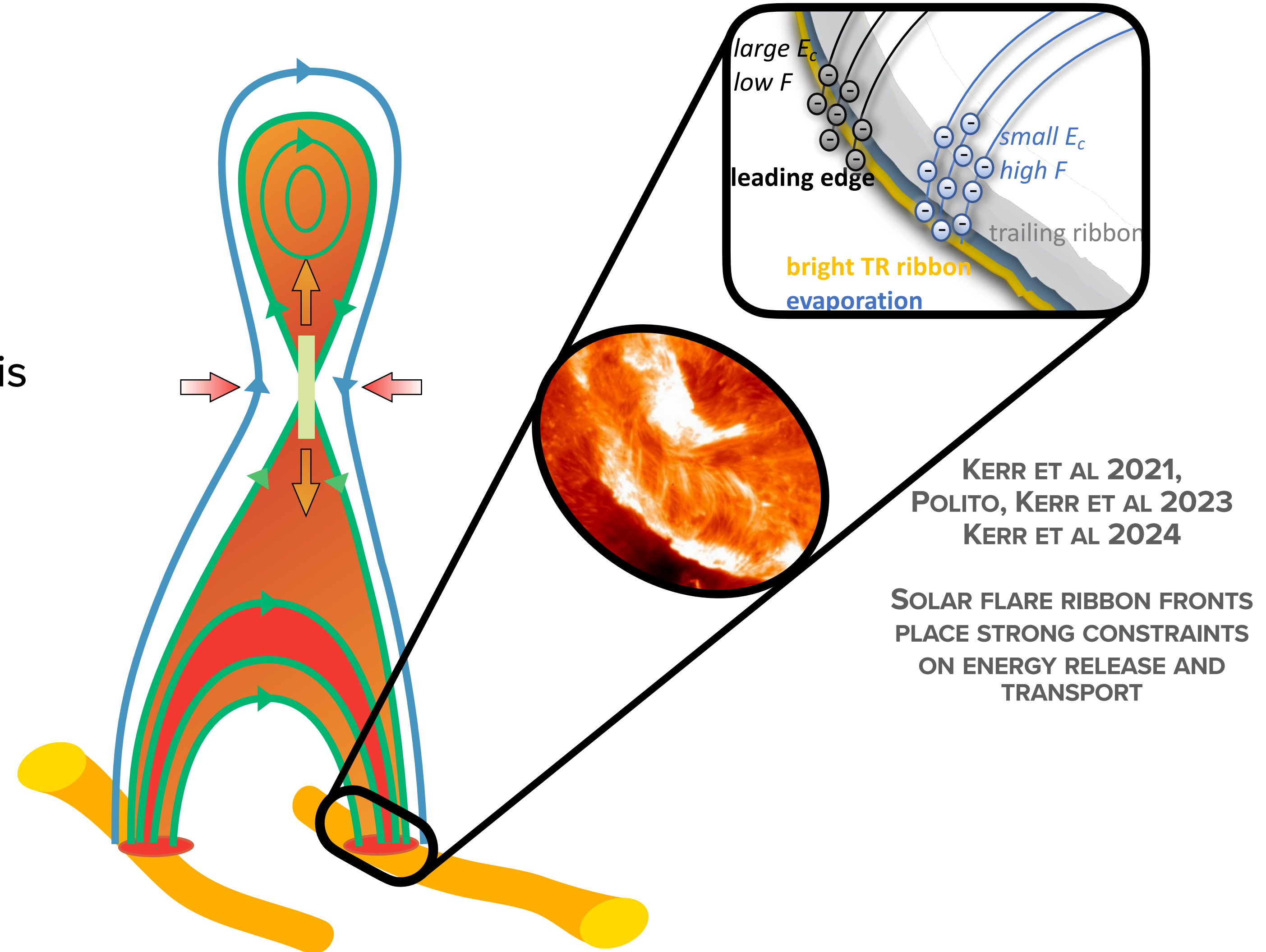
SOLAR FLARE SMALL-SCALE STRUCTURE



- Suggests that energy transport properties vary on small spatial-scales?
- *Non-thermal electron distributions.*
- *Nonthermal proton distributions.*
- *Multi-species nonthermal particle distributions.*
- *Direct In-situ heating (conductive heat flux).*
- *Flare-induced Alfvén waves (or other waves).*
- Inhomogeneous underlying chromosphere likely also plays a role!

MODELLING FLARE ENERGY TRANSPORT

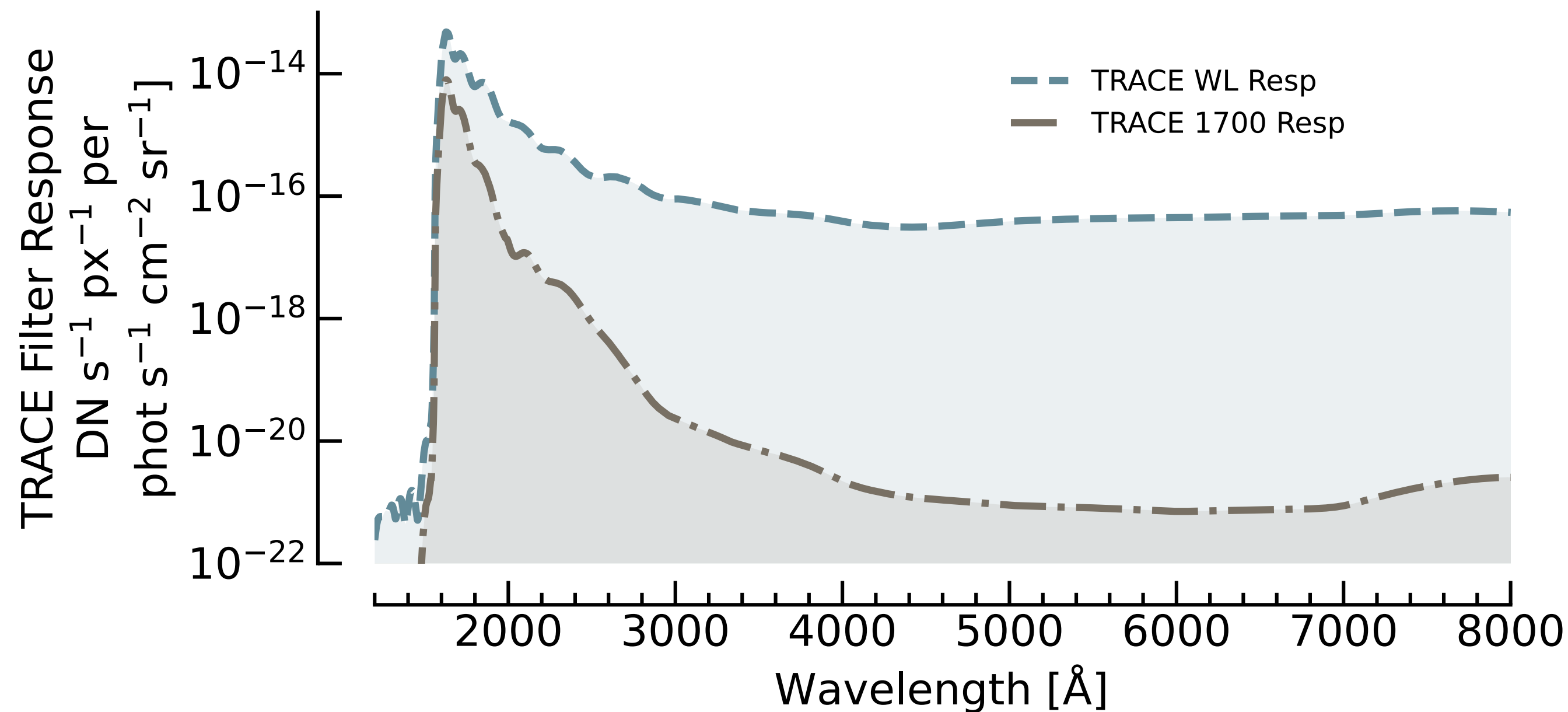
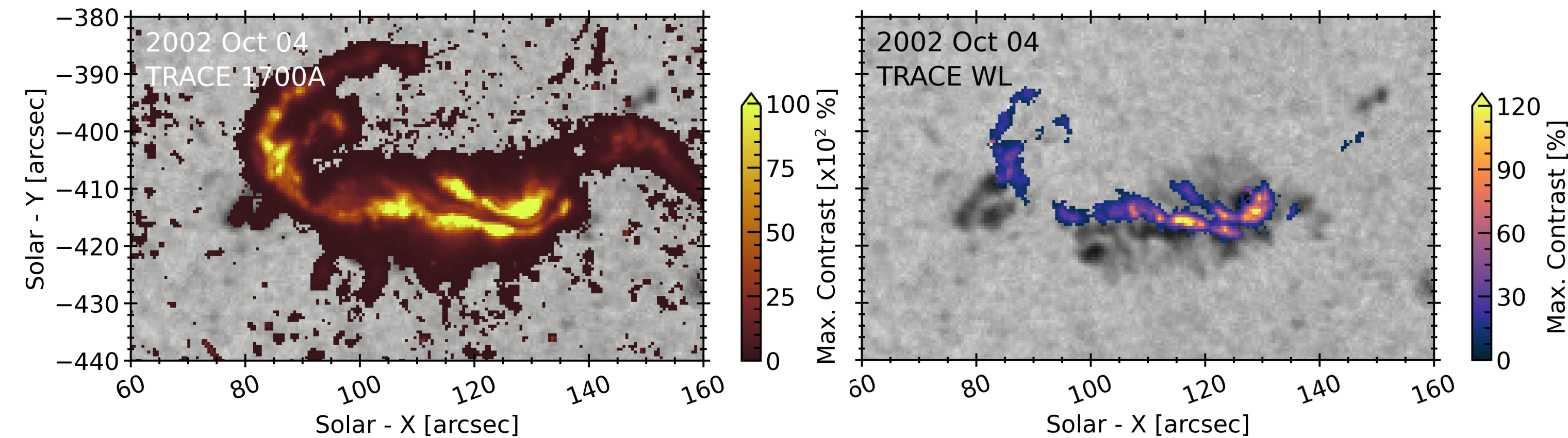
- Careful modelling has revealed much about energy transport properties and different mechanisms...
- ... However, some stubborn model-data discrepancies remain!
- How much energy is deposited, and where it is distributed in the chromosphere is still an open question.
- Knowledge of the energy flux density is key
 - ➔ Lets focus on non-thermal electrons for now... we need an accurate knowledge of the **area** into which particles precipitate.
 - ➔ White light flares (WLFs) are known to generally have more compact sources, and are closely associated with $\text{HXR} > 30 \text{ keV}$.



TRACE + RHESSI

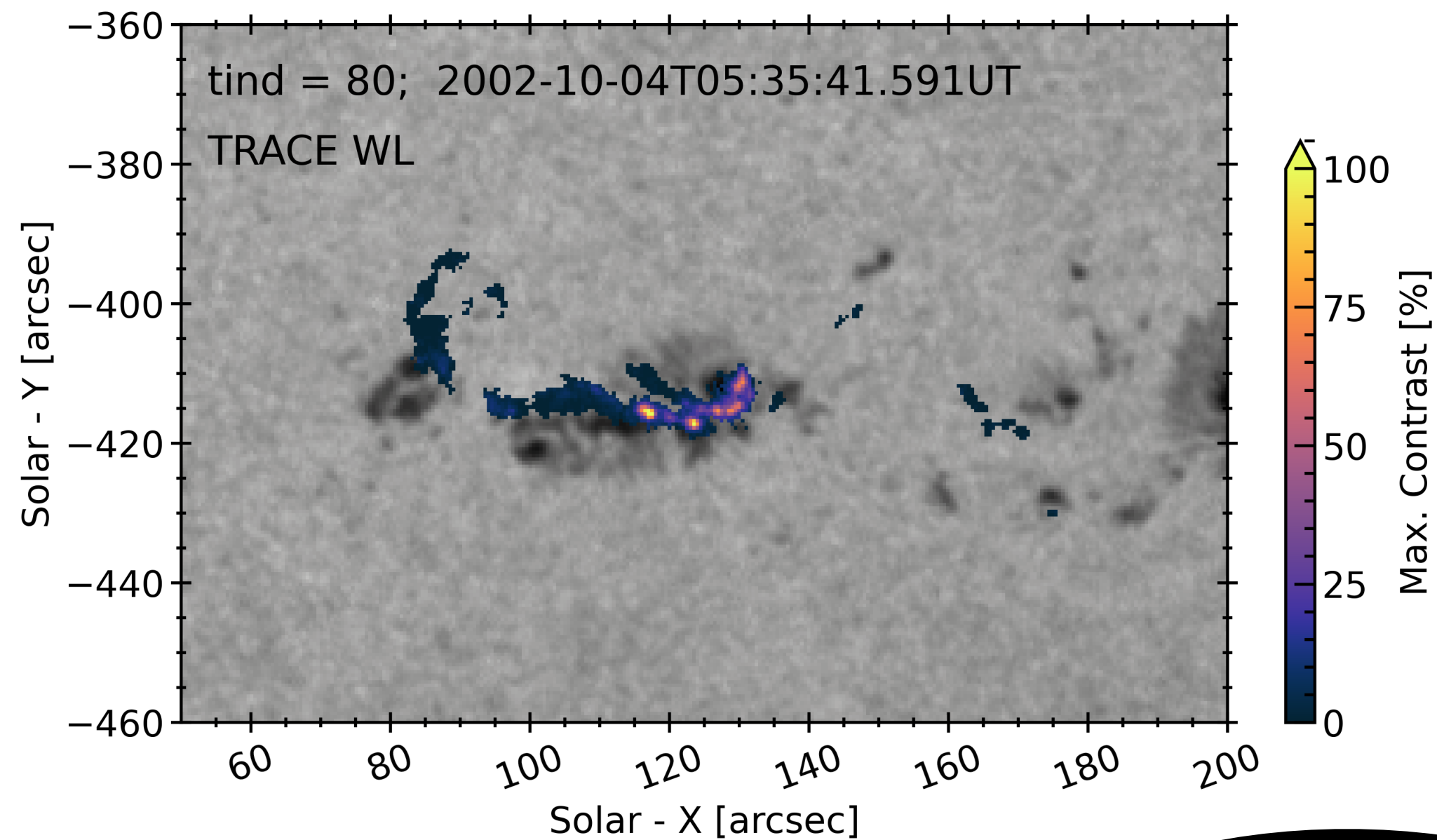
Measuring Energy Flux Density of Non-Thermal Electrons

TRACE FLARE OBSERVATIONS

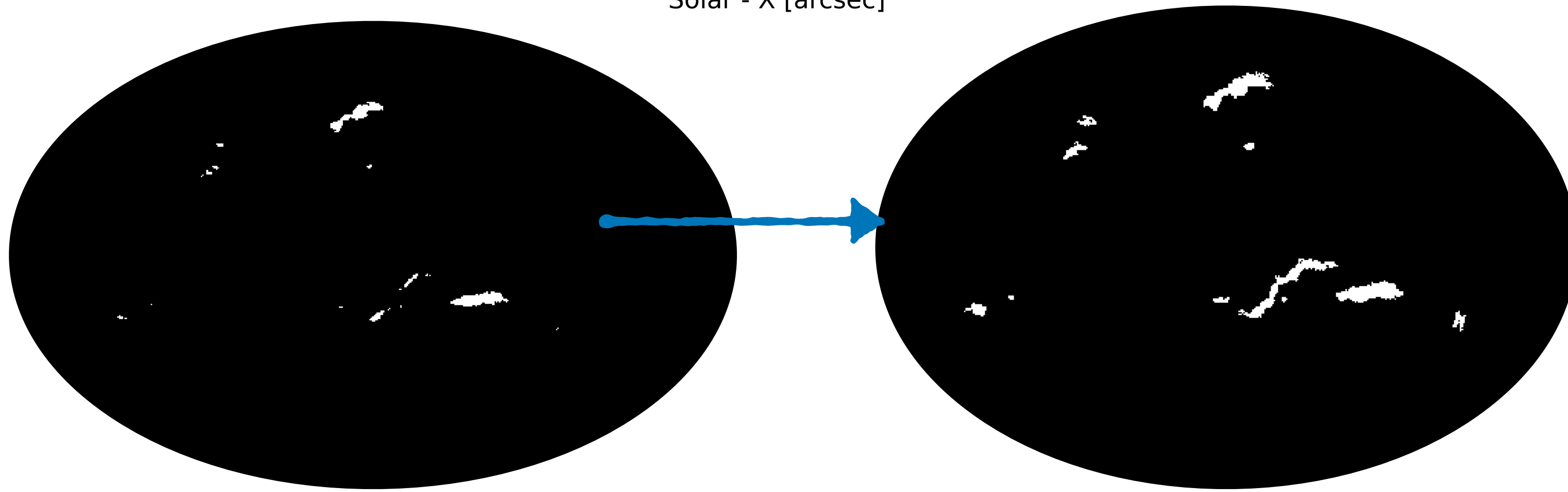


- Hard X-rays provide the power carried by non-thermal electron (NTE) distributions.
- X-ray images are unresolved — large overestimation of the area.
- UV/Optical provide much better estimates.
- We are using a series of flares observed by TRACE and RHESSI
 - ➔ Very nice data(!); we are starting by revisiting a previously analysed dataset (Fletcher et al 2007).
 - ➔ RHESSI data re-analysed, providing instantaneous measure of power in NTE in each flare.

CASE STUDY — 4TH OCT 2002

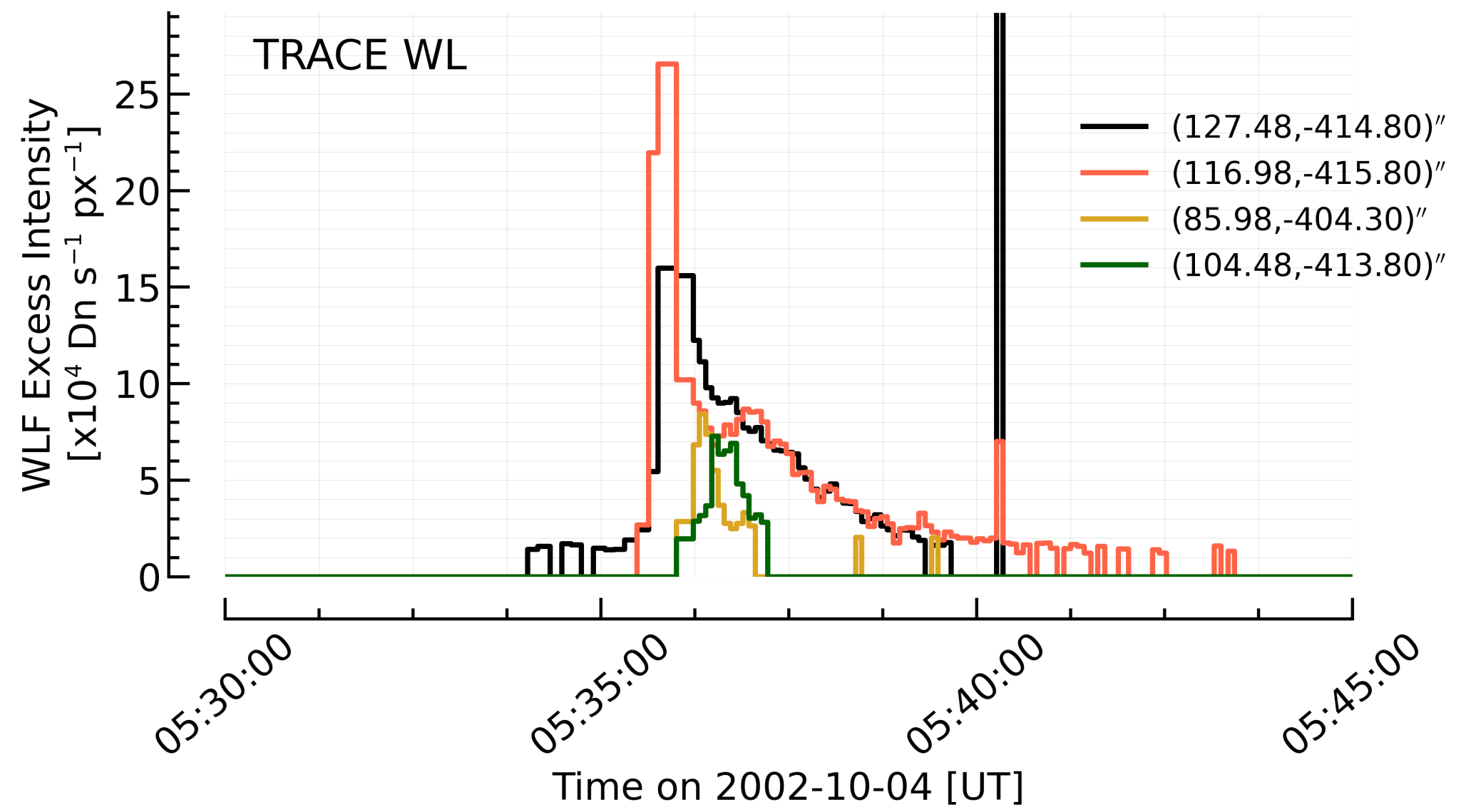
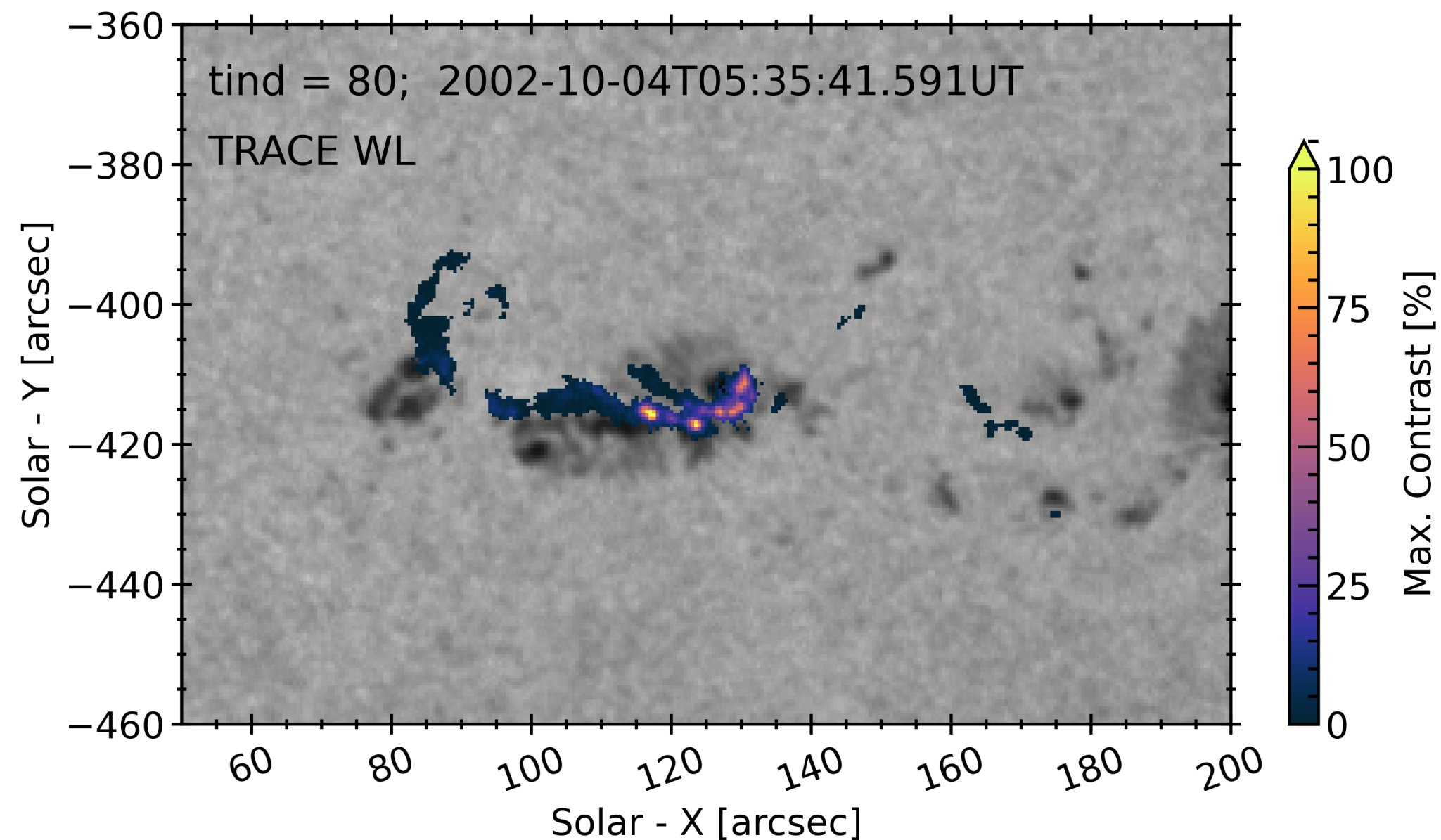


- M5.7 Class flare
- Identified WL sources though running differences where the newly brightened sources are during the time window in which the RHESSI X-rays were analyzed.
- ➔ Strict threshold to identify WL sources against bright, variable background.
- ➔ Then region growing algorithm to pick out fainter WL sources.
- Different intensity thresholds then show the range of possible areas.



REGION GROWING EXAMPLE FROM A DIFFERENT EVENT

CASE STUDY — 4TH OCT 2002



- M5.7 Class flare
- Identified WL sources though running differences where the newly brightened sources are during the time window in which the RHESSI X-rays were analyzed.
- ➔ Strict threshold to identify WL sources against bright, variable background.
- ➔ Then region growing algorithm to pick out fainter WL sources.
- Different intensity thresholds then show the range of possible areas.

CASE STUDY — 4TH OCT 2002

1700A CHANNEL AREAS

$A = 2.99 \times 10^{18} \text{ cm}^2$

$F = 8.98 \times 10^9 \text{ ERG/S/CM}^2$

ALL NEWLY BRIGHTENED WL

$A = 4.53 \times 10^{17} \text{ cm}^2$

$F = 5.89 \times 10^{10} \text{ ERG/S/CM}^2$

25% MAX NEWLY BRIGHTENED WL

$A = 1.07 \times 10^{17} \text{ cm}^2$

$F = 2.50 \times 10^{11} \text{ ERG/S/CM}^2$

50% MAX NEWLY BRIGHTENED WL

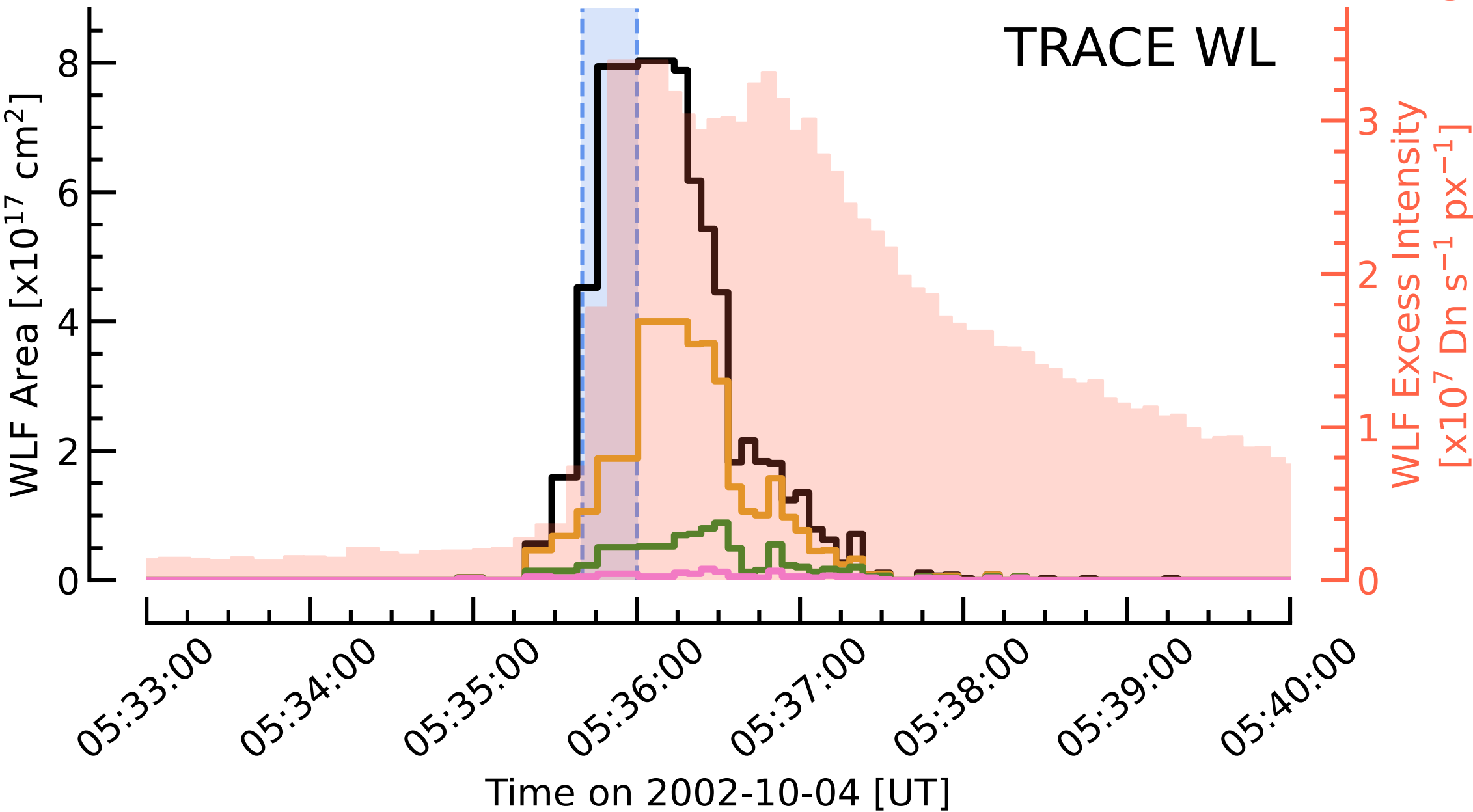
$A = 2.34 \times 10^{16} \text{ cm}^2$

$F = 1.14 \times 10^{12} \text{ ERG/S/CM}^2$

75% MAX NEWLY BRIGHTENED WL

$A = 5.84 \times 10^{15} \text{ cm}^2$

$F = 4.56 \times 10^{12} \text{ ERG/S/CM}^2$



● M5.7 Class flare

● Identified WL sources though running differences where the newly brightened sources are during the time window in which the RHESSI X-rays were analyzed.

- ➔ Strict threshold to identify WL sources against bright, variable background.
- ➔ Then region growing algorithm to pick out fainter WL sources.

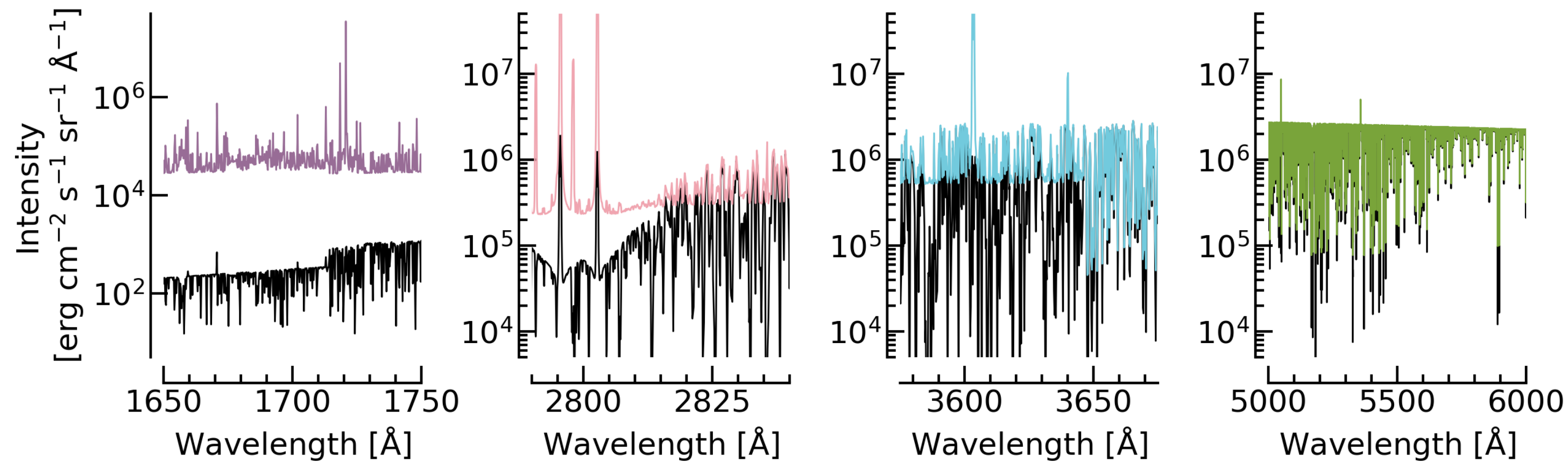
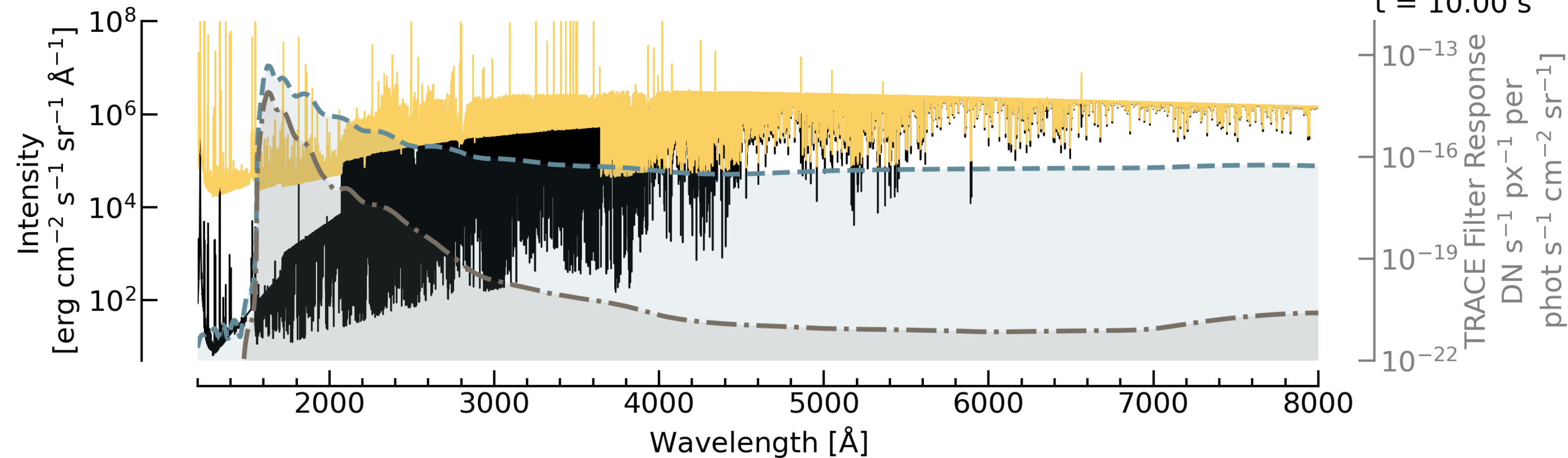
● Different intensity thresholds then show the range of possible areas.

MODELLING THE WHITE LIGHT FLARES

Brief Aside to Flare RHD Modelling

MODELLING THE FLARE 4TH OCT 2002

4th Oct 2002 -- $\delta = 5.98$, $E_c = 26.04$ keV, FI2



● RADYN + RH15D modelling.

● Four flares based on TRACE+RHESSI derived NTE parameters.

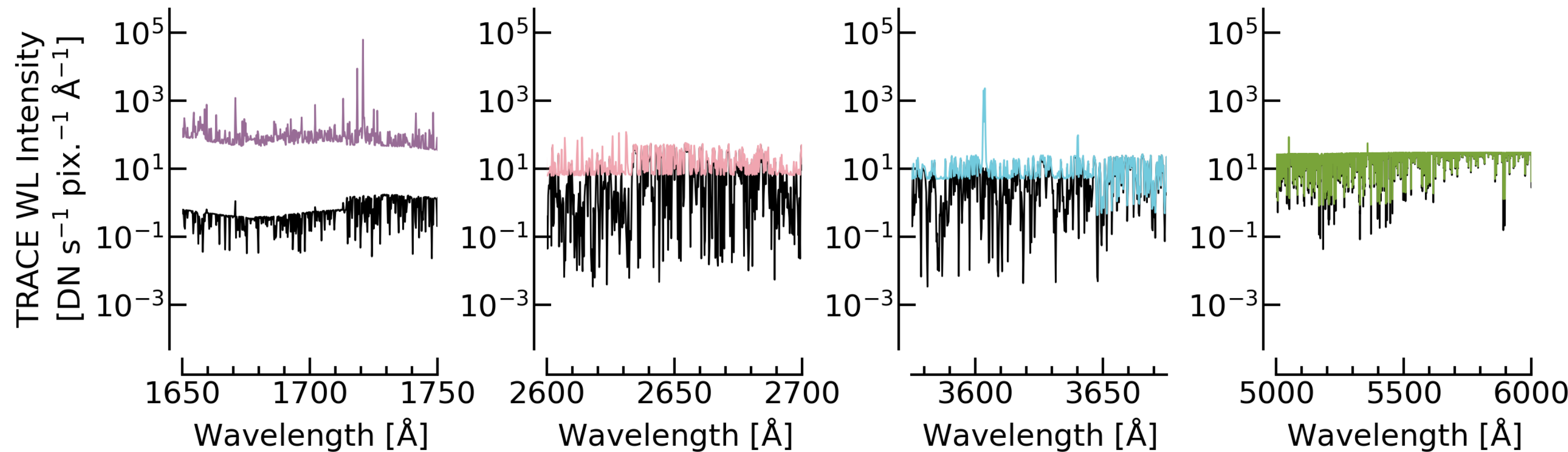
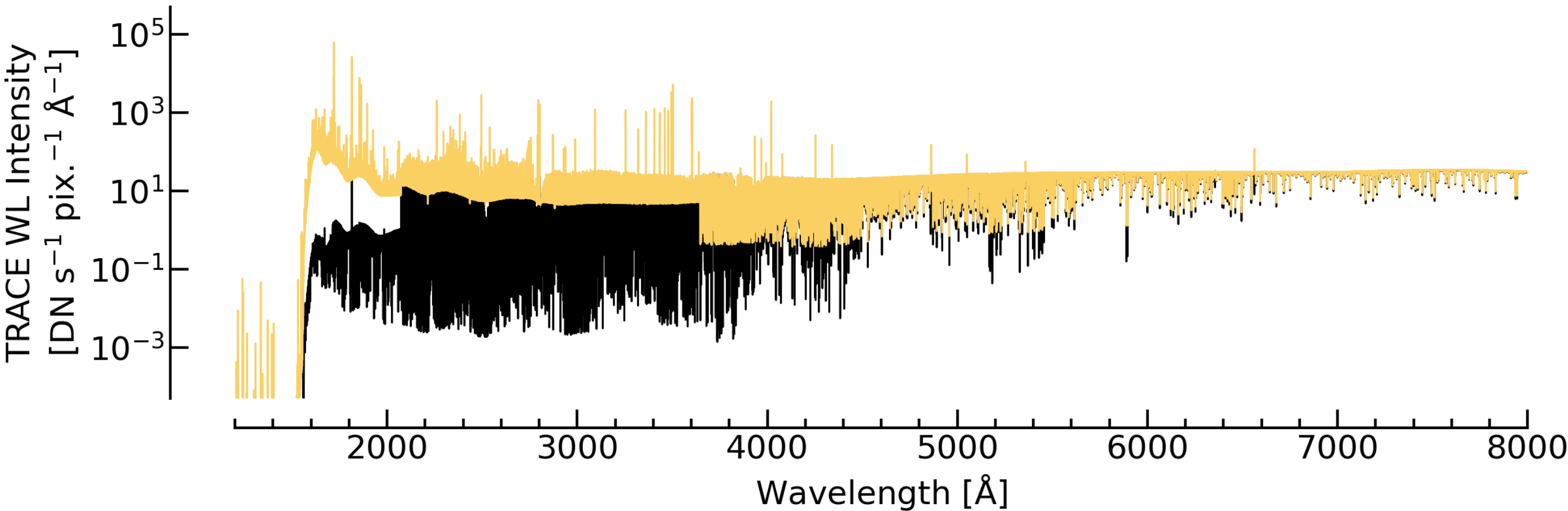
● Folded through TRACE response.

● As we would expect the large range of energy fluxes produce radically different atmospheric and radiative responses... very important to get this values as accurate as possible!

MODELLING THE FLARE 4TH OCT 2002

4th Oct 2002 -- $\delta = 5.98$, $E_c = 26.04$ keV, FI2

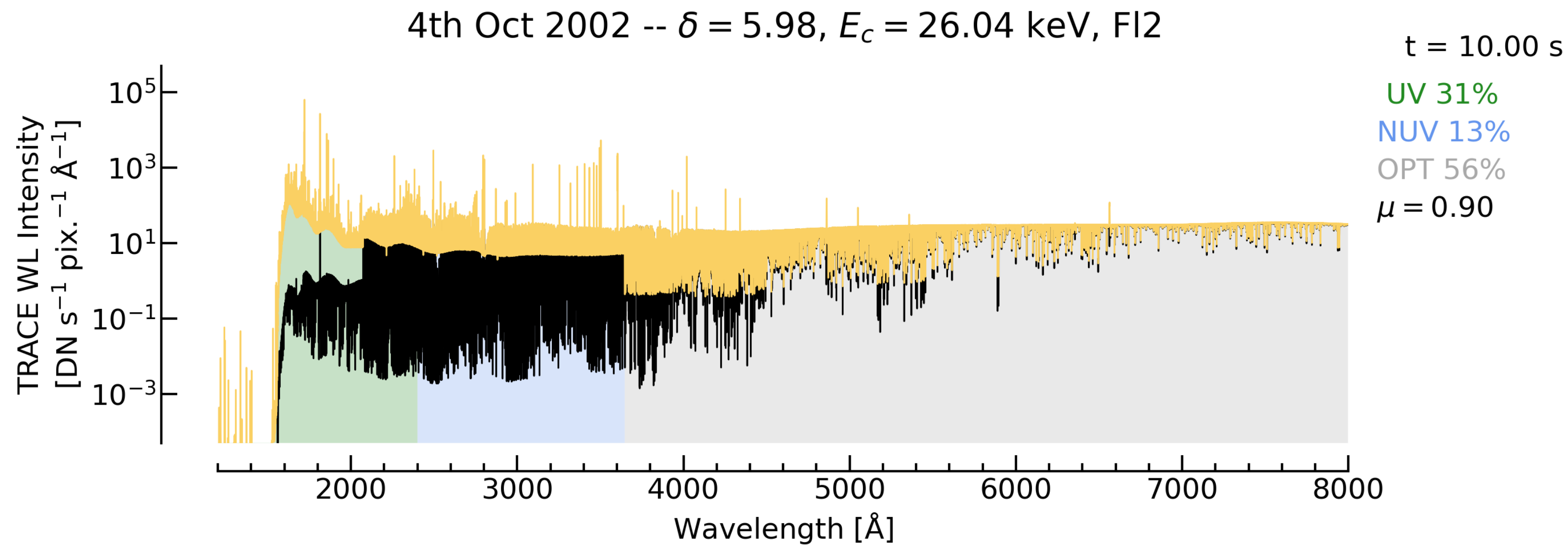
$t = 10.00$ s
 $\mu = 0.90$



FOLDED THROUGH THE TRACE RESPONSE

- RADYN + RH15D modelling.
- Four flares based on TRACE+RHESSI derived NTE parameters.
- Folded through TRACE response.
- As we would expect the large range of energy fluxes produce radically different atmospheric and radiative responses... very important to get this values as accurate as possible!

MODELLING THE FLARE 4TH OCT 2002

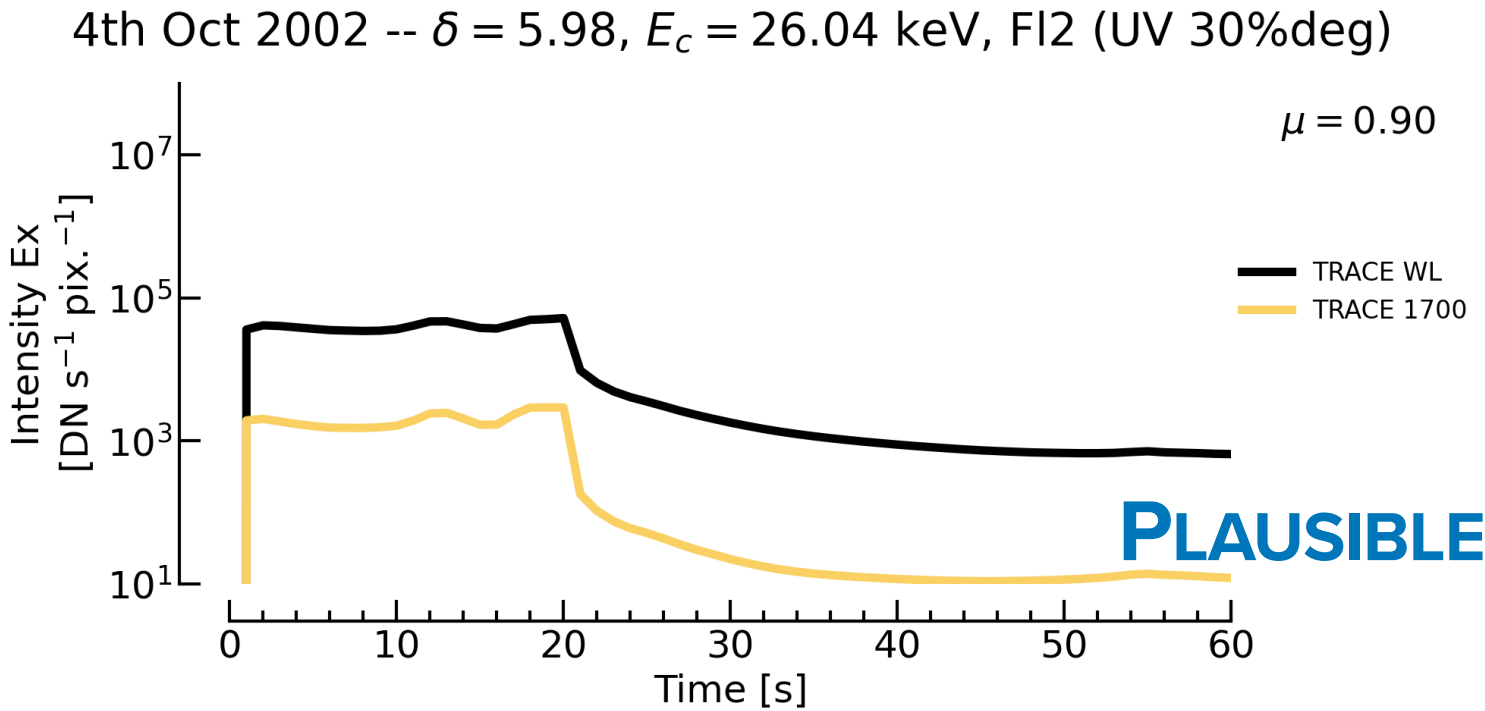


LARGE UV CONTRIBUTION TO THE WL CHANNEL!

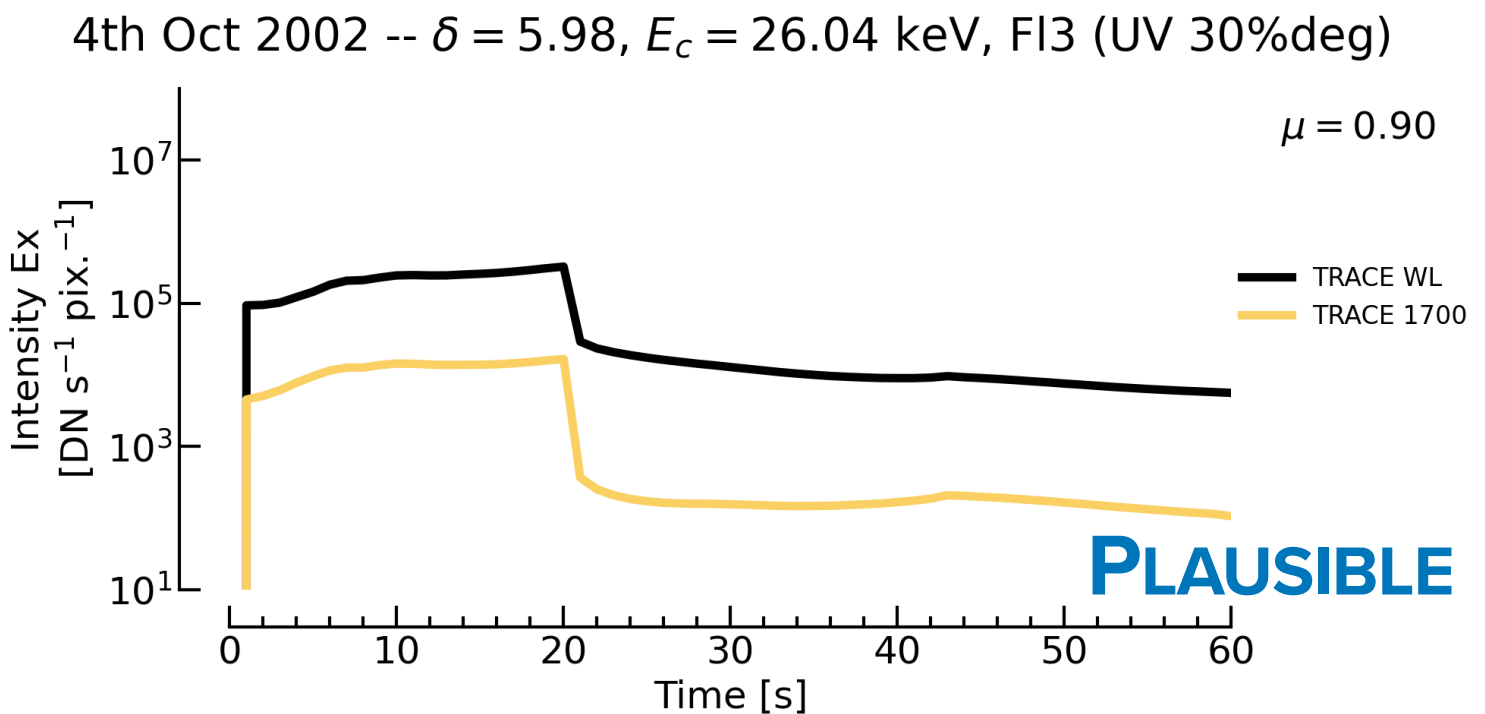
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MODELLING THE FLARE 4TH OCT 2002

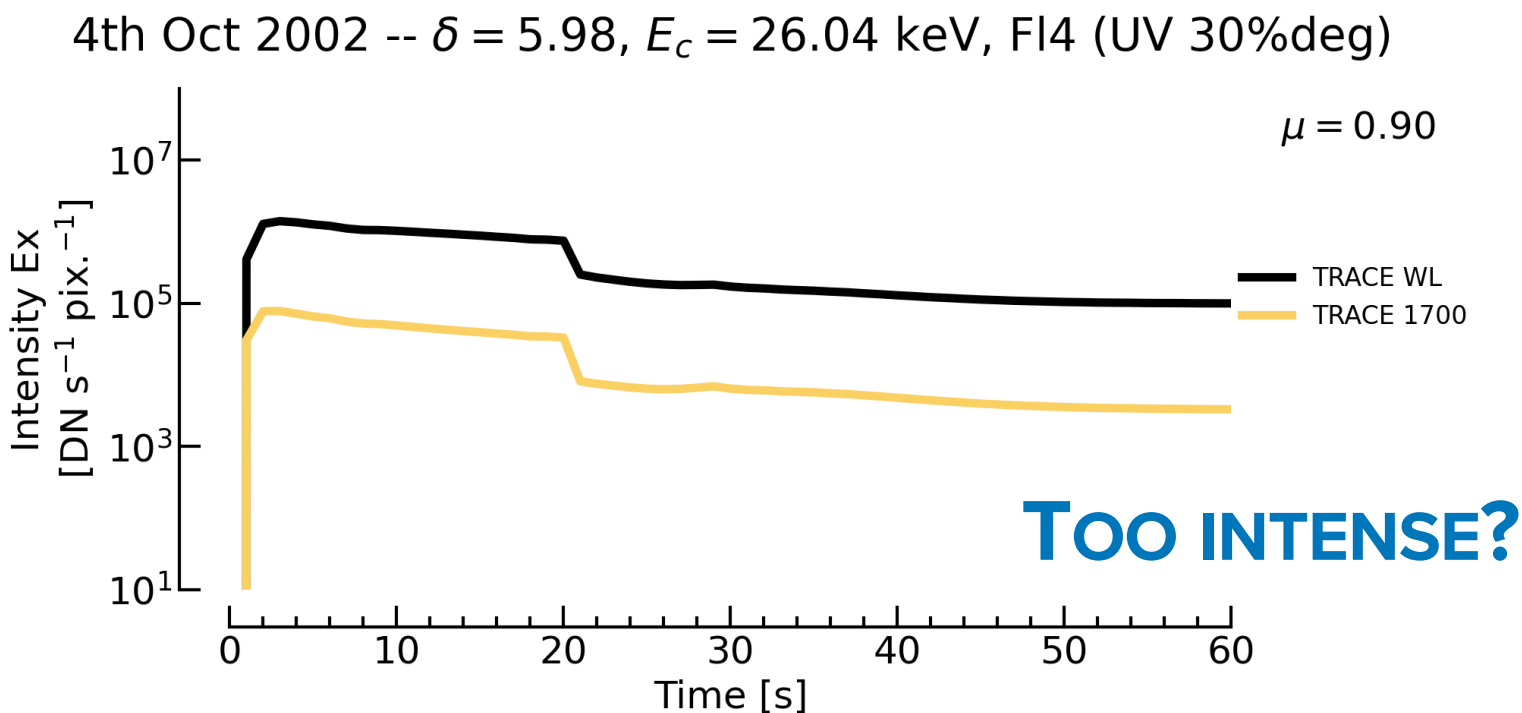
FLARE #2
 $F = 5.89 \times 10^{10} \text{ ERG/S/CM}^2$



FLARE #3
 $F = 2.50 \times 10^{11} \text{ ERG/S/CM}^2$



FLARE #4
 $F = 1.14 \times 10^{12} \text{ ERG/S/CM}^2$

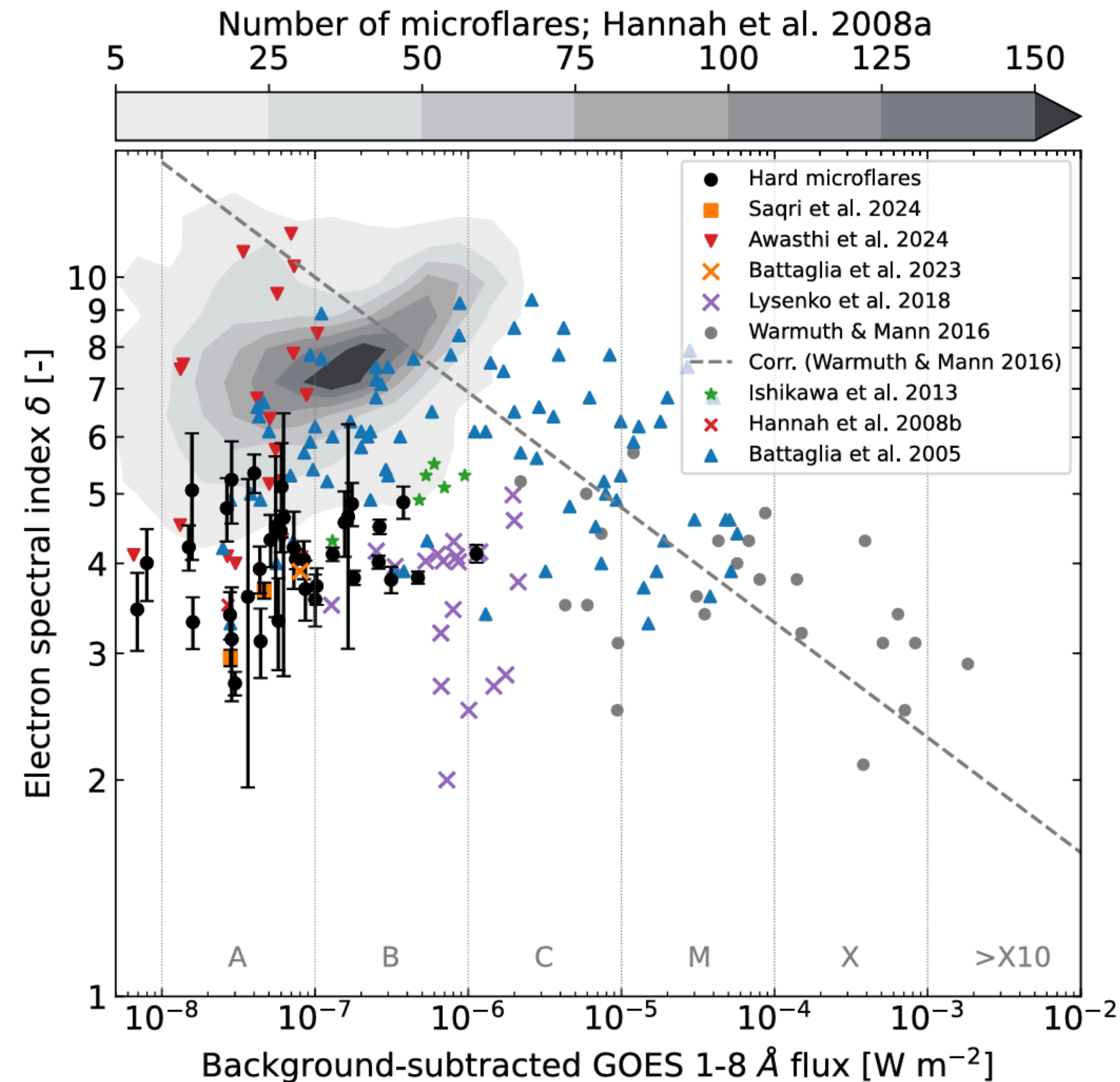


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TRACE + RHESSI

The Full Dataset

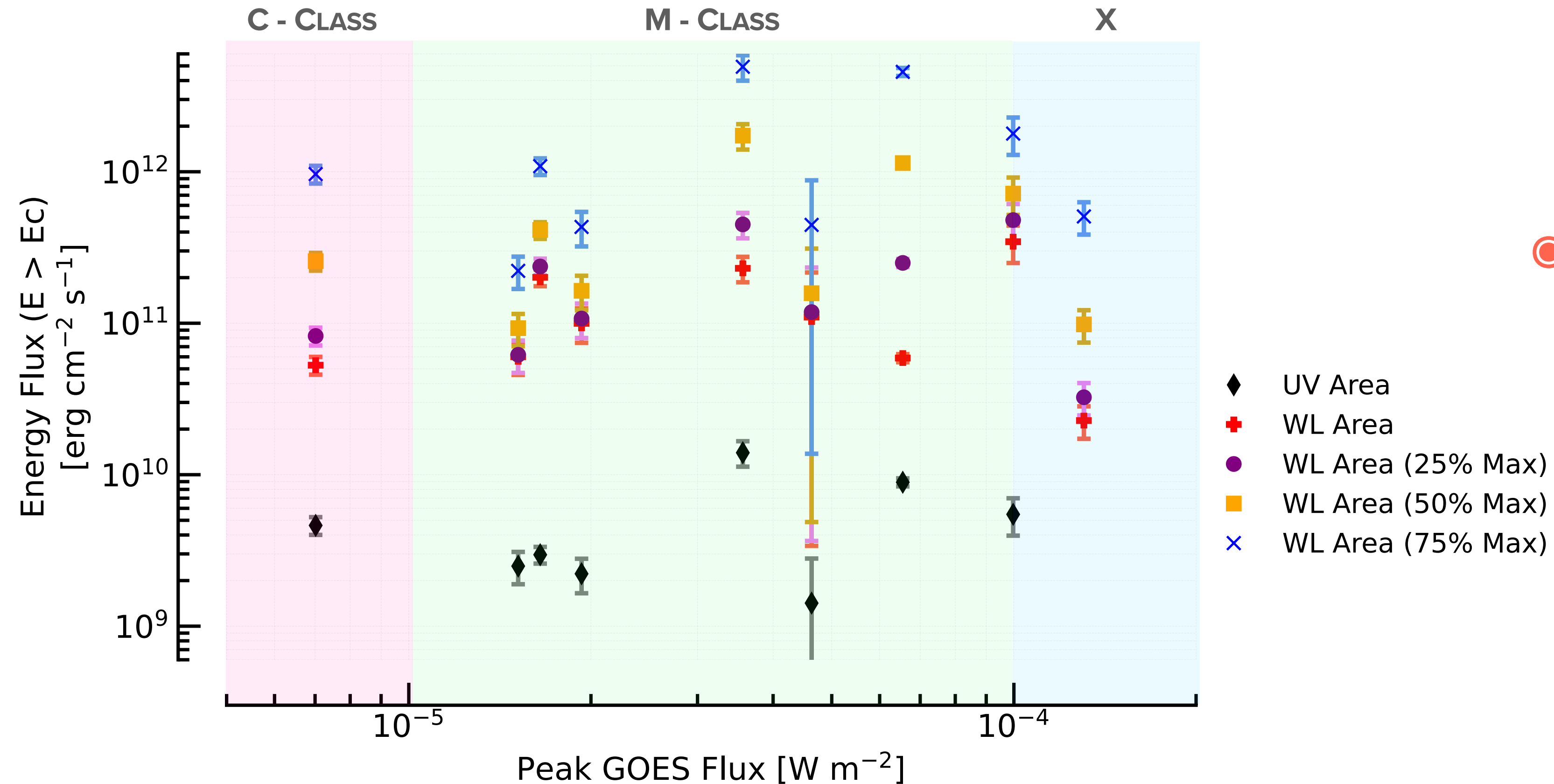
VARIATION WITH FLARE STRENGTH



- Some properties of the non-thermal electron distributions are known to vary with overall flare strength (e.g. spectral index).
- However, our careful analysis of the energy flux density suggests that there is not a strong relationship between overall flare strength in the total energy contained in the distribution!

Battaglia et al A&A 2024

VARIATION WITH FLARE STRENGTH



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However, our careful analysis of the energy flux density suggests that there is not a strong relationship between overall flare strength in the total energy contained in the distribution!

Conclusions

- The energy flux density carried by accelerated particles bombarding the lower atmosphere does not seem to vary strongly with flare strength (as measured by GOES class).
- TRACE's WL channel likely has significant UV contribution.
- Future efforts include:
 - ➔ Accounting for TRACE filter degradation.
 - ➔ Analyse recent very high-resolution Solar Orbiter EUI + STIX flares from the Major Flare Campaign.
 - ➔ Study this as function of time and space in *individual* flares.
 - ➔ **Incorporating STIX+AIA microflares from Battaglia et al 2024 to extend the range of flare classes ... how does this (lack of-) trend extend to weaker events?** (REU student @MSU, via Union College, working on this right now).

